

Portfolio Construction: From Optimisation to Hierarchy

® Abstract: This instalment examines Hierarchical Risk Parity (HRP) as a structural alternative to both Markowitz optimisation and the naïve $1/n$ benchmark. HRP avoids covariance inversion and fragile return forecasts, instead allocating capital through the hierarchy of cross-asset risk. The core question is whether diversification by structure can outperform diversification by simplicity out-of-sample.

The first two radars made it clear that the simple $1/n$ portfolio is not naïve in practice but a difficult out-of-sample hurdle. Markowitz improves the optics of optimisation, but its realised performance remains fragile because small errors in estimated returns and covariances are converted into large active bets.

HRP, introduced by López de Prado (2016), reframes the problem. Rather than asking which portfolio is optimal under noisy inputs, HRP asks which assets are genuinely distinct sources of risk. This is the key structural break from Markowitz, where weights depend on:

$$w^* \propto \Sigma^{-1} \mu.$$

The inversion of Σ is the vulnerability as it amplifies noise, concentrates capital and often rewards in-sample artefacts rather than persistent signals. The $1/n$ rule avoids this failure by ignoring forecasts altogether. It treats all holdings as equally diversifying, even when several securities are driven by the same underlying risk factor. HRP occupies a middle ground, it keeps the discipline of a rules-based allocation, but uses the correlation hierarchy for genuine diversification.

HRP begins by identifying how assets are structurally related before allocating any capital. The first step is to convert correlations into distances:

$$d_{i,j} = \sqrt{\frac{1}{2}(1 - \rho_{i,j})},$$

where highly correlated assets sit close together, while independent exposures sit further apart. This creates a hierarchical tree of risk clusters rather than a flat universe of competing securities. Once the hierarchy is established, the covariance matrix is reordered so that correlated assets sit next to each other. Capital is then allocated through recursive bisection. At each split, cluster weights are determined by relative cluster risk:

$$\alpha = 1 - \frac{V_1}{V_1 + V_2},$$

where V_1 and V_2 represent the variance of each branch. Lower-risk clusters receive more capital, while highly concentrated risk buckets are naturally scaled down. This process continues until each asset receives a final portfolio weight.

This is the structural edge of HRP. Markowitz attempts to optimise a noisy estimate of the future. The $1/n$ rule avoids that problem by ignoring structure entirely. HRP sits in the middle as it accepts that forecasts are fragile, but recognises that correlation structure contains persistent information about how risk propagates through a portfolio. It changes the topological structure of assets it optimising over.

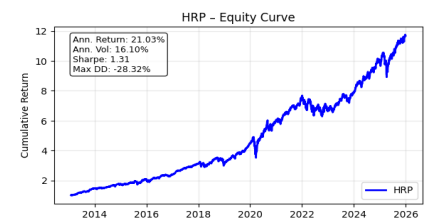
During regime shifts, correlation spikes, or crowded unwinds, portfolios built on fragile covariance inversions tend to concentrate precisely where diversification is needed most. HRP does the opposite by allocating across clusters, it naturally reduces hidden duplication of risk and produces weights that remain economically interpretable even when the covariance matrix becomes unstable.

The goal of portfolio construction is not to optimise estimates. It is to survive estimation error. The $1/n$ rule achieves this through simplicity. Markowitz often fails because it overreacts to noise. HRP establishes optimal diversification around the structure of risk itself.

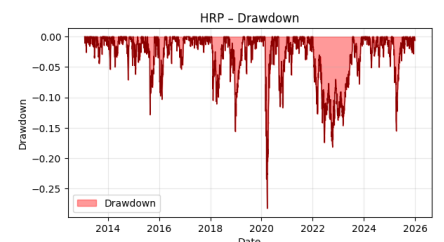
The next instalment extends the Hierarchical Risk Parity framework by incorporating the risk-reward trade-off in the allocation, without amplifying estimation error.

Charts

Out-of-sample Equity Performance



Out-of-sample Equity Drawdown



Source: prometheuscapital.nl

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